

Task 3 - Concept for an Automated Pipeline

1. Identifying Potential Issues with the Platform

To ensure stability and reliability, I would implement an automated CI/CD pipeline using GitHub Actions or Jenkins. This pipeline will run in a staging environment to detect potential issues before deployment.

Key Pipeline Stages:

1. Code Integration & Static Analysis

- Automatically pull the latest code from the QA branch.
- Run ESLint (JavaScript) or SonarQube for code quality and security checks.

2. Automated Testing

- Unit Tests for core functionalities.
- Integration & UI Tests using Playwright or Cypress.
- API Testing with Postman or REST Assured.

3. Build & Deployment to Staging

- Package the application and deploy it in a Docker container.
- Host the staging environment on AWS or a Kubernetes cluster.

4. Monitoring & Alerts

- Use Prometheus + Grafana for performance tracking.
- Set up Slack or Email alerts for failures.

5. End-to-End (E2E) Testing in Staging

- Deploy the build to a staging environment.
- Run Playwright, Selenium, or Cypress tests across browsers and mobile devices.
- Capture logs, screenshots, and videos for debugging failures.

6. Containerization & Deployment to Staging

- Create Docker containers and deploy to Kubernetes (K8s) or Docker Swarm.
- Use Helm Charts for version-controlled deployments.

2. Pipeline Execution Frequency

1. On Every Code Commit – Runs unit tests and security checks.
2. Daily Nightly Runs – Executes full regression testing.
3. Before Production Deployment – Runs acceptance tests to ensure stability.

3. Additional Applications of the Automated Pipeline

A. Assisting Large-Scale Code Refactoring

- Helps detect breaking changes early.
- Ensures backward compatibility with regression tests.

B. Supporting Developers

- Provides quick feedback on test failures.
- Automates test data generation for debugging.
- Simplifies rollbacks if deployment fails.

Conclusion

This pipeline will improve software quality and deployment efficiency by detecting issues early, automating testing, and supporting developers in their workflow.